

Bioinformatics workflow management systems

11

11.1 Introduction to workflow management systems

In this era of data-driven science, the application of bioinformatics is undergoing a fundamental transformation. Similar to the development of cosmology and the physical sciences, genomics has grown to become a significant information science. It is accompanied by a shift away from local high-performance computing (HPC) facilities and toward circulated networks and, more recently, cloud resources, particularly in large-scale multi-focus cooperative projects. In a similar way, the need to process more data at a faster rate is driving the development of software that can automate and speed up the process of looking at data in high-performance computing environments (Shade and Teal, 2015).

Scientific Workflow Management Systems, often known as WfMSs, are capable of automating computational studies by merging a variety of data-processing procedures into a single pipeline. They hide problems associated with coordinating the processing and transportation of data, managing task dependencies, and assigning resources within the computational infrastructure. There are additional WfMSs that provide techniques for monitoring the provenance of data, analyzing execution faults, authenticating users, and ensuring data security. The development of the Findable, Accessible, Interoperable, and Reproducible (FAIR) principles for scientific tools, workflows, and protocols for sharing datasets was prompted by the growing use of workflow management systems (WfMSs) in contemporary scientific research. Currently, containerized software and standard Application Programming Interfaces (APIs) are being built based on these characteristics in order to facilitate the creation, distribution, and execution of code in a variety of computer settings (Perkel, 2019).

The motivation for the development of a WfMS may have an effect on its ease of use, the functionality and features it provides, and how effectively it operates. The Common Workflow Language, often known as CWL, is a language specification that was established by the community of bioinformatics to unify the style, principles, and standards of coding pipelines in a way that was independent of the hardware being used. The repeatability of workflows and their portability are at the very top of this system's priority list. As a consequence of this, it is necessary to provide detailed and exact explanations of the parameters, which results in a long paper. Workflow

Description Language, often known as WDL, is a definition of a language that places an emphasis on making code easy for humans to comprehend and learn, but at the price of being less expressive. The combination of a workflow language and an execution engine is at the heart of the comprehensive solution known as Nextflow. It's possible that this WfMS is one of the most developed ones ever made. Nextflow has features such as easy reading, a compact size, agile, and the ability to track where the data came from. On the other hand, coding is quite easy to understand, especially for people who are just beginner at biological computing. The Swift/T system is an all-encompassing solution in the same way. Swift is the programming language that is used by Turbine, which is the execution engine. It was designed by engineers and physicists with the intention of providing short, rapid-fire tasks at exascale in a scalable way. As a consequence of this, it is a very low-level language (similar to C) that is very powerful yet challenging to master. In the following sections, we will explain how these distinct ways of thinking influence the use of the four WfMS that may be used in production bioinformatics (Leipzig, 2017).

The most important parts of a typical cloud WFMS are shown in Fig. 11.1. Creating and describing abstract workflows made up of activities and the interdependencies between them may be done using the workflow site (Ahmed et al., 2021). Reading of the abstract workflows is handled by the language parser, which is a component of the workflow enactment engine. After that, the task dispatcher investigates the connections between the different jobs and sends the finished work to the scheduler for processing. The scheduler selects a resource to do the workflow task based on the scheduling techniques that have been provided. In the next part, we will go further into the topic of task scheduling and cover it in more depth. The fault tolerance of the process is taken care of by the enactment engine for the workflow. In addition to that, it has a function for allocating resources to tasks. This is made possible through the use of the resource broker.

The resource broker is in charge of communication with the infrastructure layer, and it gives the enactment engine a comprehensive overview of everything that's going on. In order to get the necessary resources, the resource broker will interface with the computing services. Information on data objects, programs, and computer resources is stored by the directory and catalog services. This information will be used by both the enactment engine and the resource broker to come to important conclusions.

In general, workflow management services provide essential activities that are necessary for the operation of a workflow management system (WfMS). The authentication process and safe access to the WfMS are both provided by the security and identity services. Monitoring tools keep a watch on the WFMS's most important components and provide warnings whenever they deem it essential to do so. The database management component offers safe storage space for both the intermediate and final data outcomes produced by procedures. Execution information, file locations, input and output information, workflow structure, form, workflow evolution, and system information are some of the important pieces of data that are recorded by provenance management systems. Other types of data that are recorded include

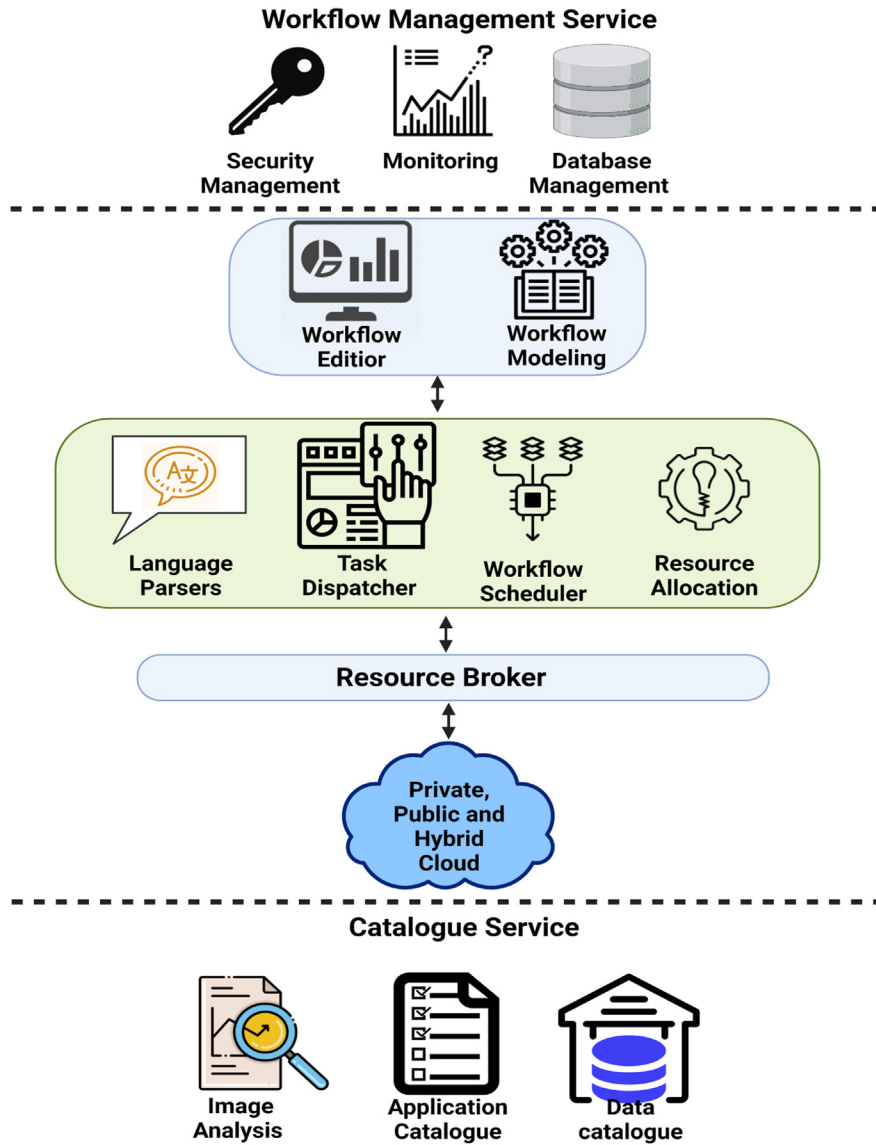


FIG. 11.1

Workflow management system.

the dynamics of control flows and data as well as how they change. Understanding the data, figuring out its quality, defining who owns it, generating repeatable outputs, maximizing efficiency, addressing concerns, and making allowances for faults are all activities that need provenance.

11.2 Galaxy

Galaxy (Cock et al., 2013) is a platform for scientific workflows that enables the integration of data, the storage of data, and analyses, as well as publishing. Research scientists who are not acquainted with programming or system administration will be able to use computational biology thanks to this project's goal of making it accessible to them. In spite of the fact that it had its beginnings in genomics research, it is today being used as a bioinformatics workflow management system for several different sorts of research.

Functionality: The Galaxy system is a way for researchers to structure their research and data. These technologies make it possible to conduct computer studies in stages, much like the steps in a recipe. They often come with a graphical user interface that gives you the ability to choose the data you want to work with, the actions you want to take, and the method you want to employ to carry out those tasks.

Galaxy also functions as a database for storing biological information. It enables users to upload data from a range of online resources, including their desktops, URLs, and other websites (such as the UCSC Genome Browser, and BioMart). Galaxy is able to interact with a wide number of conventional biological data formats as well as translate between them. Academics can format and alter text without having to learn how to code, thanks to Galaxy, which gives a web interface to a variety of text manipulation tools. Galaxy's capability for manipulating intervals may be used to carry out set-theoretic operations on the intervals that are being manipulated. The name of a chromosome or contig, as well as its beginning and ending coordinates, are examples of the types of genomic interval data that are included in a great number of biological file formats. This makes it possible to combine these pieces of data.

Galaxy was developed for the purpose of analyzing biological data, namely genomics. Over the course of time, there has been a massive expansion in the selection of tools that are at a user's disposal. Galaxy is presently used for research into a wide range of life science topics, including gene expression, genome assembly, proteomics, epigenomics, and transcriptomics, among others. Because the platform is not exclusive to any one scientific field, it is open to use in any and all scientific fields, including cheminformatics. Image classification, combinatorial chemistry, targeted therapies, cosmology, climate science, social research, and linguistics are just a few of the fields that may make use of the servers that Galaxy provides.

The Galaxy Tool Shed (<https://usegalaxy.org/toolshed>) is a central area that allows tool authors to exchange their tool settings as well as "recipes" for installing dependencies with other tool writers. This makes it easier for different Galaxy instances to share tools with each other.

Interactive analysis and visualization: Interactive analysis and visualization are two of the features offered by the Galaxy user interface, which makes it possible for anybody to do complex research. However, in order to conduct a comprehensive study of genetic data, it is often necessary to create custom scripts or visualizations, especially at the beginning (data preparation) or the end (data analysis) (data summarizing). Galaxy Interactive Environments, a Galaxy interface with Jupyter

(RStudio is in progress), a widely used interactive programming environment, was recently provided in order to meet these particular criteria. This was done so that Galaxy could fulfill these requirements. Galaxy users can make use of their current computer infrastructure by using either graphical user interfaces or ad hoc programming, or any combination of the two.

Community: As a consequence of contributions from the community, both the Galaxy framework and the collection of tools have undergone substantial development. Over the course of the last 2 years, 174 engineers have contributed to the effort to make Galaxy more scalable, functional, and user-friendly. Because of this, there have been 13,135 contributions, which is a 63% increase compared to January 2016. The project makes use of the Travis and Jenkins continuous integration (CI) services in order to do thorough automated testing on each proposed set of code modifications. This approach lowers the total number of bugs that are introduced into the codebase while also speeding up the review process.

By using the open-source community and the conventional approach to software development, the foundation is capable of releasing a stable version of the Galaxy framework every 4 months. Some of the current directions for the future development includes the following: toolshed installation and development upgrades; data and compute federation; tighter coupling of Interactive Environments with provenance and reuse; continued development of collections, processes, analysis interfaces, and historical views; more training materials; better statistical usage tracking and instrumentation; and a lot more.

Anyone who is interested in contributing to Galaxy is encouraged to read the project's contributing and code of conduct guidelines, search for outstanding issues, and review the current roadmap, all of which are available on the Galaxy GitHub repository. Galaxy has been of assistance to tens of thousands of people on a daily basis, and it has been cited in more than 5700 scientific works. In addition to this, it has provided over 500 developers with a framework that makes the process of data analysis straightforward, observable, and re-useable. Galaxy Main (and more than 99 other public Galaxy servers) have been set up in order to make it easier to conduct research in many fields, including biological research and study in other domains. During the last 2 years, the Galaxy Project has made great headway in all fields, which has led to overall improvement (Fig. 11.2).

11.3 Gene pattern

GenePattern (Kuehn et al., 2008) is a piece of computational biology software that was created at the Broad Institute and is available for free and open-source use. It is used for the processing the genomic data. GenePattern was originally made available to the public in 2004, and its primary purpose was to provide academics with a platform on which they could construct, store, and implement genomic analysis algorithms. GenePattern is currently being developed by researchers at the University of California, San Diego.

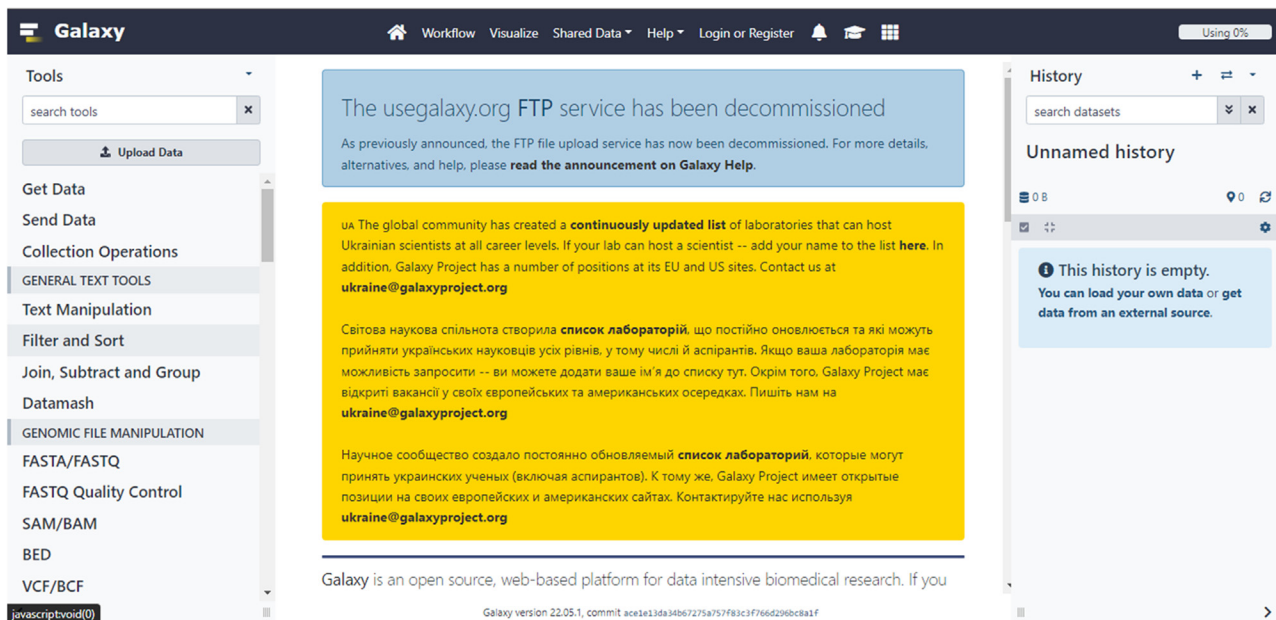


FIG. 11.2

Galaxy interface.

GenePattern is a repository of hundreds of bioinformatics analysis and visualization techniques (referred to as “modules”). It also contains utilities for data formatting, preprocessing, and other functions that serve as “glue” between analysis steps. Simply pointing and clicking will get you through the user interface; no coding is required. Since 2008, individuals have been authorized to use the public GenePattern server, which can be found at www.genepattern.org. It has a user base of around 40,000 people and does between 2000 and 5000 analytical activities each week. Public servers are also accessible at Indiana University and the Garvan Institute. More than 17,000 research groups, bioinformatics core centers, and individual scientists have downloaded and used the software.

GenePattern is software that operates on both a client and server level. Components of an application may be operated on a single computer with requirements as low as those of a laptop, or they can be distributed over several computers, allowing the server to make use of hardware with a higher processing capacity. The GenePattern server is really the GenePattern engine. It runs the analysis modules and stores the results of those analyses when they are generated. Both the Web Client and the Desktop Client are examples of graphical user interfaces that simplify the process of accessing the server and all of the modules that it contains. A web browser is required in order to access the Web Client once it has been installed concurrently with the server. The desktop client is an independent piece of software that must be downloaded and installed on each individual machine. GenePattern libraries for the Java, MATLAB, and R programming environments allow you to access the server and its modules via function calls in addition to providing access to those environments. The essential protocols described in this part are carried out with the help of the Web client. However, these protocols may also be carried out with the assistance of the desktop client or a programming environment (Fig. 11.3).

The majority of transcription profiling studies aim to accomplish at least one of the following goals: investigate differences in gene expression; find new classes; or anticipate future classes. The goal of differential expression analysis is to discover (if any) genes that are expressed differently in distinct groups or kinds of samples. This may be done by comparing the gene expression levels of a number of different groups or types of samples. Genes that express themselves in unique ways are referred to as marker genes, and the process that identifies these genes is referred to as marker selection. The process of merging genes or samples that have similar expression profiles into a smaller number of patterns or “classes” is known as class discovery. This process offers a high-level view of the microarray data. It is possible to identify common biological processes by categorizing genes according to the degree to which their expression patterns are comparable to one another. It is possible to identify common biological states or subtypes of diseases by grouping samples according to the degree to which their expression patterns are comparable to one another. There are a number of different ways that one may use data on gene expression to find classes. The goal of research into class prediction is to locate crucial marker genes whose patterns of expression can correctly categorize unlabeled data.

**FIG. 11.3**

GenePattern interface.

- **Accessibility:** Utilize a point-and-click user interface to run over 200 analysis and visualization tools, such as data preprocessing, gene expression analysis, proteomics, single nucleotide polymorphism (SNP) analysis, flow cytometry, and next-generation sequencing, and develop analytical workflows without the need for scripting.
- **Reproducibility:** Any user is able to understand, duplicate, and share the results of a whole computational research thanks to versioning, automated history tracking, and provenance monitoring.
- **Extensibility:** The computational approach, users are able to collaborate and share their processes and code with one another by using tools that simplify the generation and combination of such elements.

11.4 KNIME: The Konstanz information miner

The KNIME Analytics Platform ([Fillbrunn et al., 2017](#)) is open-source software for data science, which means that anybody is free to use it. Anyone may do data analysis and construct data science processes and reusable components with the help of KNIME since it is user-friendly, open-source, and continually updated with new features. The following is a list of some of the features that are included in KNIME Builds Workflows: Using a simple drag-and-drop user interface, it is possible to create visual workflows even if you lack writing skills. The user has the ability to construct workflows, model each step of an inquiry, manage how data flows, and guarantee that the work is up-to-date by picking from over 2000 nodes that are provided by KNIME. The software also incorporates other technologies, such as machine learning, Python or R programming, or Apache Spark connections, into a single workflow. This is done by integrating KNIME native nodes with these extra technologies ([Fig. 11.4](#)).

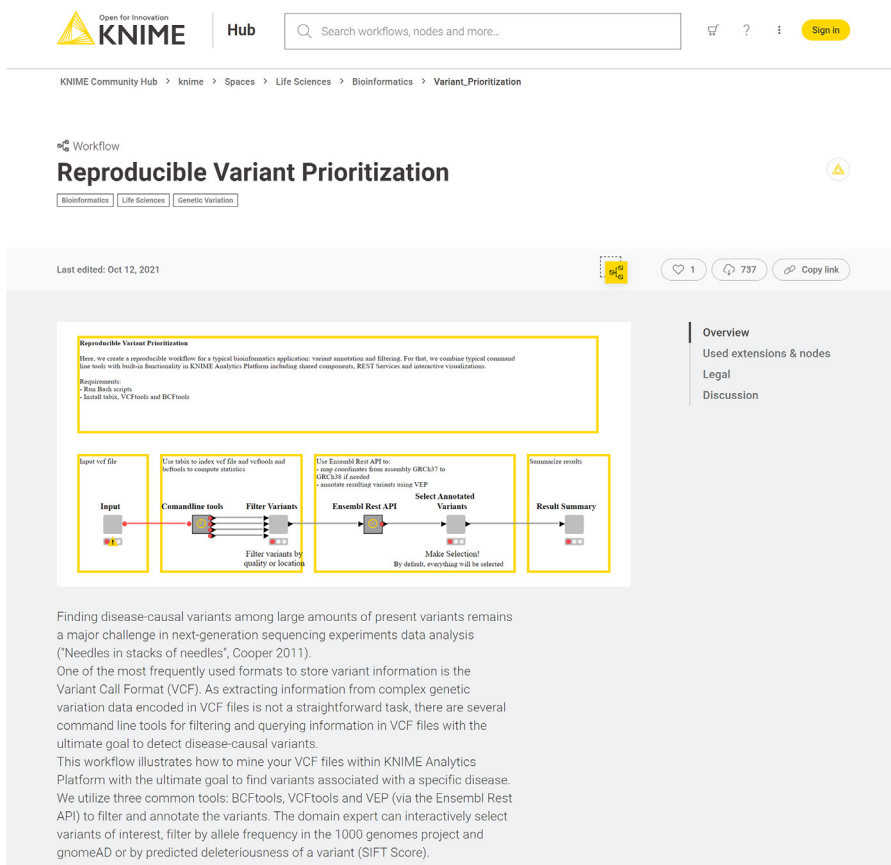


FIG. 11.4

Knime variant prioritization workflow.

Data that is combined: KNIME allows for the mixing of a variety of data kinds and formats, including structured and unstructured data types, as well as time-series data. Some examples of these formats are PDF, CSV, XLS, and JSON. It establishes connections to Oracle, Microsoft SQL, Apache Hive, and other databases and data warehouses so that it may aggregate data from those sources. Data comes from many different places, such as Twitter, AWS S3, Google Sheets, and Azure.

Data on Shapes: performs calculations using statistics such as the mean, quantiles, and standard deviation, or uses statistical tests in order to establish a hypothesis. Workflows could also incorporate other elements like correlation analysis, cutting down on the number of dimensions, and other similar things. Data can be collected, sorted, filtered, and combined on a single workstation, in a database, or in a “big data” environment where the data is spread out over several workstations and databases.

Machine Learning and Artificial Intelligence: provides machine learning models for classification, regression, dimension reduction, or clustering utilizing advanced methodologies such as deep learning, tree-based algorithms, and logistic regression. These models may be used to analyze large amounts of data. Increasing the performance of a model may be accomplished in a number of different ways, such as by adjusting the hyperparameters, boosting, bagging, stacking, or generating sophisticated ensembles. Performance measures like accuracy, R2, the area under the curve (AUC), and ROC are used to determine how well a model functions. Cross-validation is a method that may be used to check whether or not a model is reliable. For the purpose of describing machine learning models, LIME and Shapley values are used. The interactive presentation of partial dependency and ICE helps the tool have a better understanding of the model's predictions and make forecasts based on tried-and-true models or the most advanced PMML available, including Apache Spark.

Insights to Discover and Share: The data is presented via user-customizable charts that range from straightforward (bar charts, scatter plots) to intricate (parallel coordinates, sunburst, network graph, heat map). This command will remove any extraneous information from a KNIME table and show a summary of the statistics for each column. In order for stakeholders to see the results, reports may be exported in a variety of formats, including PDF, PowerPoint, and others. The processed data or the results of the analysis are often saved in a file format or database that is widely used.

Scale Execution in Response to Demands: develops workflow prototypes in order to allow for the testing of several different approaches to data analysis. Examine and monitor the intermediate results in order to get rapid feedback and discover original ideas. The speed of the workflow may be increased by processing data using several threads and streaming data in memory. Increases the speed of calculations by using either the database processing capabilities of Apache Spark or its distributed computing capabilities.

KNIME Extensions provide access to strong machine learning algorithms as well as a diverse selection of complex data types. KNIME Analytics Platform and KNIME Server collaborate with many other open-source projects. Access data from Apache Hadoop and systems that store Hadoop data, such as Hive and Impala. You may model and execute Apache Spark jobs in local KNIME configurations to leverage the potential of scalable analytics. Create a model for generating predictions, then apply it to fresh data, or just use R or Python code inside a KNIME process to create various forms of visualization. Deep neural networks may be read, created, edited, trained, and executed.

HiLiting: KNIME is strongly reliant on the HiLiting system. It gives the user the ability to choose and highlight a number of rows inside a data table, and those rows are then highlighted in all of the other views that show the same data table (or at least the high-lighted rows). Due to the one-to-one relationship that exists between the tables' distinct row keys, it is not difficult to implement this sort of highlighting. There are numerous nodes that change the structure of the input table, but the rows that are

linked between the input table and the output table do not change. The techniques used for clustering data are an excellent illustration of a 1:n link. One of the inputs that the node receives is the training (or test) patterns, and one of the outputs that it produces is the cluster prototypes. Each cluster is in charge of its own unique collection of pattern inputs. When one or more clusters in the output table are highlighted, any input patterns in the input table that relate to those highlighted clusters are also highlighted. It is possible that other summarizing models, such as the branches and leaves of a decision tree, common patterns, and discriminative molecular fragments, may be translated in the same way.

11.5 LINCS tools

The LINCS project ([Xie et al., 2022](#)) was established on the idea that if any one of the several steps of a biological process is altered, the molecular and cellular features, behavior, and/or function of the cell will be altered. This idea serves as the project's guiding principle. The term for this characteristic is "cellular phenotype." By watching how and when different inputs change the phenotype of a cell, scientists may be able to understand more about the underlying mechanisms that lead to chaos and, ultimately, illness.

A number of data releases were carried out in order to make the LINCS data available to the general public as a community resource. Because of this, scientists are able to work on a wide array of basic research problems, which will make it easier to discover biological targets for innovative disease treatments. The LINCS databases include the findings of tests conducted on cultured human cells as well as primary human cells that were treated with bioactive small compounds, ligands such as growth factors and cytokines, or genetic changes. These tests were performed on human cells. Assays that measure transcript and protein expression, in addition to biochemical and imaging readouts that collect data on cell phenotypic characteristics, are used so that the response of cells may be monitored.

The LINCS project is a two-part study funded by the NIH Common Fund. 2013 was the last year of the pilot phase of the program, which was made up of the following parts:

- Large-scale signatures of the molecular and cellular modifications that were produced by the disruption.
- In order to provide users with access to data, databases, data standards, and public user interfaces are now being developed.
- Developing new computer programs and gathering information.
- Recent advances in inexpensive molecular and cellular phenotyping

11.5.1 The program's overall goal

By employing an integrative strategy that searches for patterns of common networks and cellular responses (termed "cellular signatures") across various types of tissues

and cells in response to a variety of changes, LINCS is attempting to find a novel approach to understanding health and illness. This is being done in order to advance the mission of the organization. The LINCS program is established on the idea that changing any one of the many stages that make up a biological process may result in changes to the molecular and cellular characteristics, behavior, and/or function of a cell. This concept serves as the program's guiding principle. After that, the phenotypic state of a cell might be represented by signatures that were acquired through similar clinical tests. By examining how and when different stimuli change the phenotypic characteristics of a cell, scientists may be able to learn more about the systems that are involved in disruption and, ultimately, sickness.

A collection of data descriptors with decreased dimensionality that may both shed light on a process and serve as a basis for making predictions is what is meant to be understood as a cellular signature of a response to a perturbation. As a consequence of this, significant fingerprints are established based on the assay and the manner in which a number of tests are combined, either into prediction patterns or signaling networks that may lead to mechanistic explanations. The results of many tests need to be normalized and compared before they can be considered relevant. Integration, normalization, and scaling of heterogeneous, multi-parameter dose-response data is a difficult process that requires both theoretical and practical considerations.

11.5.2 Test performed under LINCS

The strategy calls for the establishment of six Data and Signature Generation Centers. The Drug Toxicity Signature Generation Center is comprised of a number of different centers, including the HMS LINCS Center, the LINCS Center for Transcriptomics, the LINCS Proteomic Characterization Center for Signaling and Epigenetics, the MEP LINCS Center, and the NeuroLINCS Center, among others. The Drug Toxicity Signature Generation Center developed assays that measure gene and protein expression in addition to phenotypic characteristics in order to gain a better understanding of how differentiated iPSCs react to single and multiple FDA-approved medication modifications. This was done in order to learn more about how differentiated iPSCs respond to the changes.

The HMS LINCS Center employs a number of different biochemical, imaging, and cell biology tests in order to track the responses of cells. Imaging experiments, transcriptional response tests (done in partnership with the LINCS Center for Transcriptomics), cell viability assays, and direct drug-kinase interactions in cell extracts are some of the others.

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FIG. 11.5
Lincs interface.

11.6 Anduril bioinformatics and image analysis

Anduril (Cervera et al., 2019) is a tool for doing process analysis on massive amounts of data. It comes with ready-to-use tools for conducting analyses in the field of molecular biology, which may also be put to use for a variety of other tasks. Anduril makes it easy for people who use analytics to see the results, and it gives analysts a strong workflow environment in which to work.

The iterative workflow engine is used to run workflows in parallel once they have been created using Scala 2.11, which is used to generate the workflows. External libraries (like R and Python) and tools run from the command line are sometimes included in workflows. Anduril may be set up on a single computer or over an entire network of Linux computers and devices.

Software Development: The vast majority of the most famous bioinformatics frameworks, such as Anduril 2, are able to handle serial and parallel processes, intricate dependencies, a wide range of software and data file formats, as well as user-defined parameters and deliverables. Anduril presents the connections between components in the form of a graph and may automatically combine elements that do not depend on one another in any way. Because of the universal prefixing of the processes, SLURM and Sun Grid Engine may be used in a wide variety of different ways.

Re-entrancy: When running lengthy and complicated pipelines on enormous datasets, it is to the user's advantage to be able to resume execution from the point where it was stopped. This prevents the user from having to figure out where to start again or which samples have already been processed, which is a time-consuming and laborious task. You are free to make changes to the component settings or add new samples to the process at any point without the unfinished independent stages being forced to run. As a direct result of this, much less time is needed for both the calculation and the programming.

Dependency support: Any change to a step, such as an adjustment to a parameter, will cause all of the processes that are dependent on that step to begin over from the beginning. It is possible to annotate the inputs and outputs of a component in order to create artificial connections between them in cases where they are not physically linked. A component, for example, may not have any outputs, but it could still be able to change its environment in some way, such as by altering a database, which would then trigger the execution of another component on which it depends.

Resources in bioinformatics: There are around 400 well-described parts and functions that may be used to carry out standard tasks for a variety of bioinformatics research projects. These components can be used. The bulk of third-party software that is supported by Anduril components comes with its own installer, which makes it a great deal easier to install the many software packages that are necessary for standard bioinformatics research. Anduril 2 may utilize either its own installation or one that the user has specified on their end. Additionally, since the effective configuration of each component is stored in a bash script, any component may

be operated outside of Anduril 2 with the same settings and pipeline inputs, provided it has the same script. Testing is made easier as a result.

Ease of integrating new tools or custom analysis: It is not difficult to include additional tools in a pipeline since eval-based components may be integrated with either their own code or the code of a third party. To add a new tool to the component repository for either private or public use, all you need to do is define its inputs, parameters, and outputs in an XML file, preferably together with test cases. After that, the tool will be available for use. For example, Taverna necessitates the use of third-party software in order to develop plug-ins that may be integrated into pipelines. Both Galaxy 2 and Anduril 2 make it easy to develop wrappers; however, Anduril also makes it easy to incorporate custom analysis and software into any pipeline. Galaxy 2 is the more recent of the two.

11.6.1 Anduril image analysis: ANIMA

The modern microscope generates an enormous quantity of picture data, which then has to be analyzed and interpreted using techniques specific to computers. In addition, a single image analysis project may need tens or hundreds of steps of analysis, beginning with the entry of data and pre-processing, moving on to segmentation and statistical analysis, and concluding with the visualization and reporting of the results. Anima is a workflow framework that is modular in nature and is used for organizing large-scale picture data processing projects.

Digital Image Analysis: The use of high-tech equipment such as X-ray machines, microscopes, and MRI and CT imaging equipment to produce digital photographs has been an integral part of our day-to-day lives for quite some time. This includes the taking of digital photographs with smartphones and their subsequent sharing on social media. The production and transmission of semantic information from digital images are referred to as “digital image analysis,” and it is a technique that is in great demand across a variety of industries and applications.

The Anima workflow system, which is an acronym that stands for ANduril IM-age Analysis, is a modular system that gives us the ability to analyze images in a quick and comprehensive manner. By enabling batch processing, Anima has made high-throughput image analysis very easier to do.

Processing, interoperability, and the freedom to run on any platform are just some of the perks that come along with this. Anima is mainly geared toward developers of algorithmic and analytic tools, and it gives these developers the ability to combine a wide variety of computing tools into a single workflow system. The primary objective of Anima is to make Rapid Application Development (RAD) possible as well as the incorporation of new methods without the need of porting them from their original implementations. Regardless of the computer language that is being used, Anima gives researchers in the field of image analysis the ability to experiment with novel methods and integrate their findings into established procedures.

Anima's architecture has a fundamental flaw in that it was designed to be used as a platform for enhancing the functionality of other apps rather than as a replacement for such programmes in their entirety. Anima has proven to be a useful tool for a variety of tasks, including the creation and evaluation of innovative algorithms as well as the execution of routine analyses. It has a horizontal data flow architecture, which means that each step, like picture segmentation, can be done at the same time on all of the images.

Horizontal flow management gives you the ability to make certain that each stage of processing yields the desired results before moving on to the next phase. Because Anima is a platform that is adaptable, extensible, and scalable, it is very easy to add new features to the system while also guaranteeing that those features may be used effectively several times. You are also able to quantify attributes based on all of the images using Anima, which you can then use in the process of analysis. Anima is a really powerful tool.

11.7 NextFlow

Nextflow ([di Tommaso et al., 2017](#)) is a reactive workflow architecture and domain-specific language (DSL) that was developed at the Barcelona Center for Genomic Regulation by the Comparative Bioinformatics Group (CRG). It lets scientists use software containers to make scientific procedures that can be scaled up and used over and over again.

Nextflow may be run either locally or on a dedicated instance of the Amazon Elastic Compute Cloud. If your procedures require a significant amount of computing resources, go with the second choice. However, after your procedure is over, you are required to terminate the instance. The architecture that is shown here explains how to make the most of AWS Batch in order to run Nextflow in a way that is both simple and economical to administer.

An RNA-Seq analysis may be carried out by following a sequence of steps known as the nextflow-core RNA-Seq workflow. This process was developed by the community. The DNA sequence data will first be converted into the raw FASTQ file format before undergoing quality control, alignment to the reference genome, quantification, and differential expression calculation. Over the course of its development, this pipeline has seen over 3700 separate changes, all of which can be seen on GitHub. The procedure, by default, consists of 20 steps and uses a number of different programs. When compared to starting from scratch, using this strategy will provide results far more quickly.

Workflows made with Nextflow should be able to handle millions of samples as long as enough computing resources are available. For instance, 23 and Me uses NextFlow to handle the genetic data of its customers. However, when biological data is converted into the structure and size that are generally handled by data engineering procedures, bioinformatics workflow managers may not be the best answer. Ginkgo Bioworks is a genomics company, whose pipeline processes terabytes of

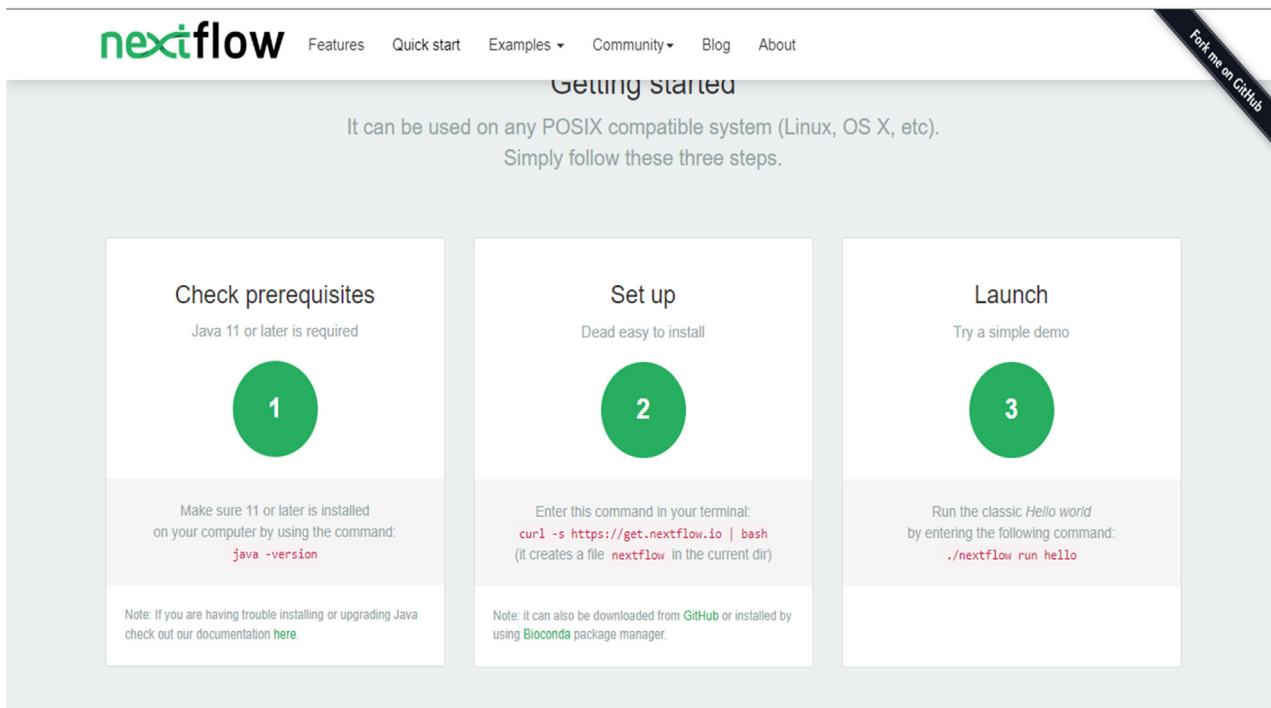


FIG. 11.6

NextFlow interface.

sequencing data every single day. Ginkgo uses an Airflow, Celery, and Amazon Web Services batch process. At industry scale, efficiency is of the utmost importance, and the solution that Ginkgo has developed was developed by an entire team of data engineers. The majority of biotechnology companies as well as university labs would benefit more from using Nextflow or another bioinformatics-specific workflow management system that may be set up by a single scientist.

The capability of isolating the workflow implementation, which specifies the flow of data and the operations that are to be done on that data, from the configuration parameters that are needed by the execution platform is one of the most important aspects of Nextflow. This makes the process portable, so it can run on many different computer platforms, like an institution's high-performance computing (HPC) system or a cloud architecture, without having to change the way the workflow is designed (Fig. 11.6).

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